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Open-Source AI Coding Tools: Comprehensive Architecture Guide

Last updated: March 25, 2026 **Scope:** All major open-source AI coding CLIs, IDE extensions, editors, and agent platforms

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Comparison Table

Tool	Category	Stars	License	Language	Models	MCP	Active	Last Commit
OpenCode	CLI	130,096	MIT	TypeScript	75+ (all major)	Yes	Very Active	Mar 25, 2026
Gemini CLI	CLI	99,036	Apache-2.0	TypeScript	Gemini family	Yes	Very Active	Mar 25, 2026
OpenHands	Agent Platform	69,738	MIT	Python	All major LLMs	Yes	Very Active	Mar 25, 2026
Cline	IDE Extension	59,338	Apache-2.0	TypeScript	All major + local	Yes	Very Active	Mar 25, 2026
Aider	CLI	42,366	Apache-2.0	Python	All via LiteLLM	No	Very Active	Mar 17, 2026
Tabby	Self-Hosted	33,050	Apache-2.0	Rust	CodeLlama, Star-Coder, etc.	No	Active	Mar 2, 2026
Goose	CLI/Desktop	33,554	Apache-2.0	Rust	All major LLMs	Yes	Very Active	Mar 25, 2026
Continue	IDE Extension	32,047	Apache-2.0	TypeScript	All major + local	Yes	Very Active	Mar 25, 2026
Void	Full IDE	28,469	Apache-2.0	TypeScript	All major + local	No	Paused	Jan 12, 2026
Roo Code	IDE Extension	22,830	Apache-2.0	TypeScript	All major + local	Yes	Very Active	Mar 19, 2026
bolt.diy	Agent Platform	19,204	MIT	TypeScript	19+ providers	Yes	Active	Feb 5, 2026
SWE-agent	CLI/Research	18,844	MIT	Python	GPT-4o, Claude, etc.	No	Active	Mar 24, 2026
Kilo Code	IDE Extension	17,177	MIT	TypeScript	All major + local	Yes	Very Active	Mar 25, 2026
bolt.new	Agent Platform	16,265	MIT	TypeScript	Anthropic (hosted)	No	Moderate	Dec 2024

Tool	Category	Stars	License	Language	Models	MCP	Active	Last Commit
Sweep AI	IDE Plugin	7,655	Proprietary (OSS core)	Jupyter/Python	Own LLMs	No	Active	Sep 2025
Cody	IDE Extension	3,794	Apache-2.0	TypeScript	Claude, GPT-4o, Gemini	No	Pivoted	Mar 24, 2026
Devon	Agent Platform	3,454	AGPL-3.0	Python	Anthropic, OpenAI, Groq	No	Stale	Jul 2024
Mentat	CLI	2,559	Apache-2.0	Python	OpenAI, Anthropic	No	Archived	2024
CodeGeeX	IDE Plugin	7,593	Apache-2.0	Python	CodeGeeX (13B)	No	Low	2025
Pear AI	Full IDE	670	MIT	TypeScript	GPT-4, Claude, Llama	No	Low	May 2025

1. CLI Tools

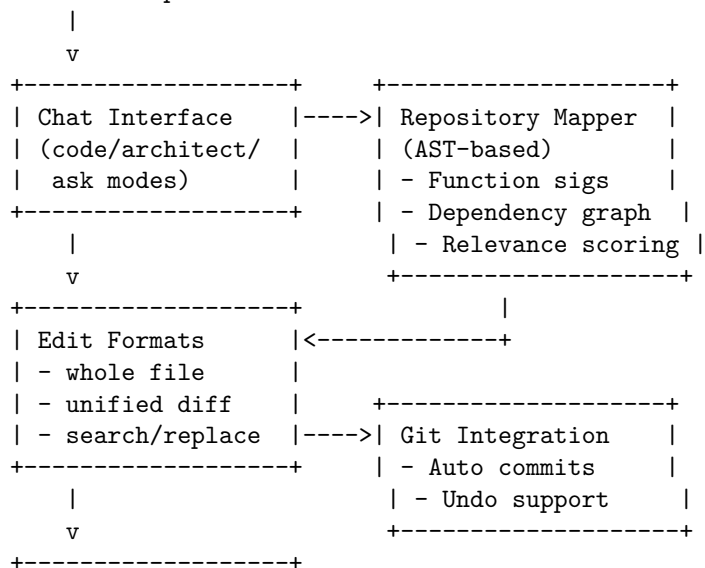
1.1 Aider

Field	Value
Repo	github.com/Aider-AI/aider
Stars	42,366
License	Apache-2.0
Language	Python
Created by	Paul Gauthier
Latest commit	Mar 17, 2026

What it does: AI pair programming in the terminal. Aider lets you collaboratively edit code with LLMs by adding files to a chat session, making multi-file edits, and auto-committing changes with descriptive git messages.

Architecture:

Terminal Input



```

| Lint & Test Loop |
| - Auto-fix errors |
| - Re-run tests   |
+-----+

```

Key architectural innovations: - **Repository Map:** Uses Abstract Syntax Trees (AST) to create a structural map of the entire codebase – function signatures, class hierarchies, import relationships. This map is used to prioritize which context to send to the LLM. - **Architect/Editor dual-model system:** An “Architect” model (e.g., Claude Sonnet) designs the solution, then an “Editor” model translates that into precise code edits. Each model specializes in its strength. - **Multiple edit formats:** Supports whole-file replacement, unified diffs, and search/replace blocks. The format is auto-selected based on the model’s capabilities. - **Auto lint & test:** After every edit, Aider runs your linter and test suite, then automatically attempts to fix any failures.

Models supported: All major LLMs via LiteLLM – OpenAI (GPT-4o, o1, o3-mini), Anthropic (Claude Sonnet 4, Opus 4), Google (Gemini 2.5), DeepSeek (R1, V3), xAI, Azure, OpenRouter, Bedrock, Vertex AI, local models via Ollama. Works best with Claude Sonnet 4 and DeepSeek.

Key features: - Multi-file coordinated edits across 100+ programming languages - Voice-to-code via microphone input - In-chat commands (/add, /drop, /undo, /diff, /test, /lint) - Git integration with auto-commits and undo - Architect mode for planning before coding - Watch mode for continuous development - Web content scraping for context - `.aider.conf.yml` for project-specific configuration

Strengths: - Most mature CLI coding tool in the ecosystem (started 2023) - Excellent multi-file editing with structural awareness - Model-agnostic with deep LiteLLM integration - Strong SWE-bench scores, regularly benchmarked - Clean git history with descriptive auto-commits - Active solo maintainer with frequent releases

Weaknesses: - No MCP support (as of March 2026) - No built-in browser automation - Python-only (slower startup than Go/Rust alternatives) - No persistent sessions (restarts lose context) - No native subagent/parallel execution - UI is basic terminal text (no TUI)

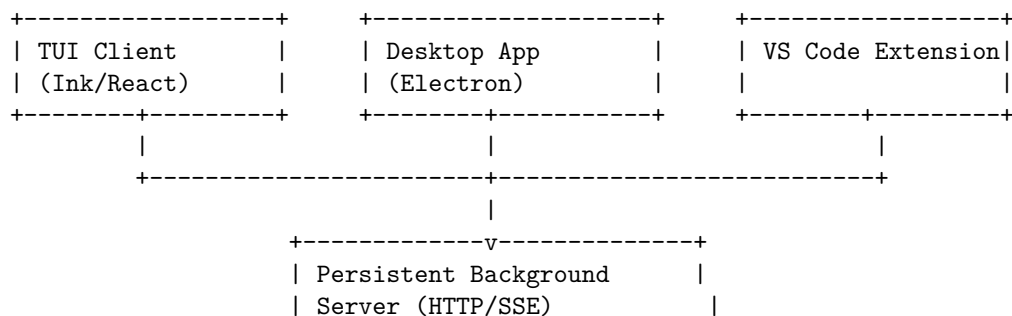
Community: 42K+ stars, 4,068 forks, 1,455 open issues. Very active solo-maintainer project with strong community contributions.

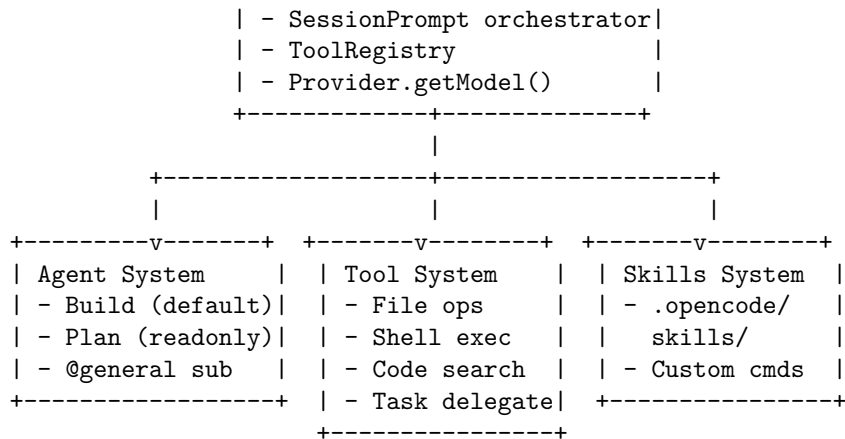
1.2 OpenCode (sst)

Field	Value
Repo	github.com/sst/opencode
Stars	130,096
License	MIT
Language	TypeScript
Created by	SST (Anomaly)
Latest commit	Mar 25, 2026

What it does: The most-starred open-source coding agent. A terminal-native AI coding tool with a rich TUI, persistent sessions, multi-agent system, and support for 75+ LLMs across all major providers.

Architecture:





Key architectural innovations:

- **Persistent background server:** Sessions survive terminal disconnects, SSH drops, and machine sleeps. You reconnect and pick up where you left off.
- **Multi-interface:** Same backend serves TUI (terminal), desktop app (Electron), VS Code extension, and web interface.
- **7 native agents:** Build, Plan, and specialized subagents with predefined permissions and tool access. Agents can delegate to other agents.
- **Layered tool system:** Tools from various sources registered in ToolRegistry, resolved per-agent and per-model capabilities, transformed for the AI model by ProviderTransform.
- **Skills system:** First-class skill definitions in .opencode/skills/ that integrate with the execution pipeline.

Models supported: 75+ models – OpenAI, Anthropic, Google Gemini, AWS Bedrock, Azure OpenAI, Groq, OpenRouter, local models via Ollama and LM Studio, and essentially any OpenAI-compatible API.

Key features:

- Rich TUI with session management
- Persistent sessions that survive disconnects
- MCP server support for extensibility
- Skills system for reusable prompts/workflows
- Custom commands via Markdown files
- LSP integration for code intelligence
- Privacy-first (no code/context data stored by OpenCode)
- File operations, shell execution, code search built-in

Strengths:

- Largest open-source coding agent by stars (130K+)
- Multi-interface (CLI, desktop, IDE extension, web)
- Persistent sessions are a killer feature for long tasks
- Massive model support (75+)
- Very active development and large community (800+ contributors)
- Privacy-focused architecture

Weaknesses:

- Relatively new (viral growth in late 2025 / early 2026)
- Still pre-1.0 in many interfaces
- TypeScript-based (heavier than Go/Rust alternatives)
- Community documentation still catching up to rapid feature development

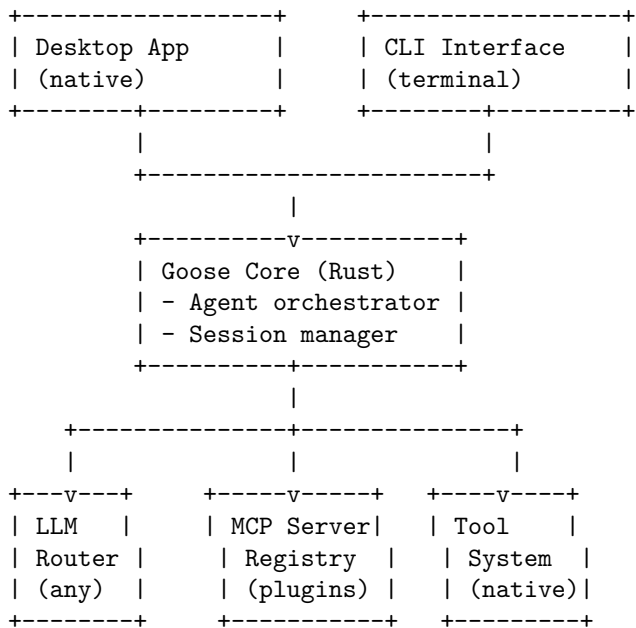
Community: 130K+ stars, 13,793 forks, 800+ contributors, 10K+ commits, 5M+ monthly developers. The fastest-growing open-source coding project in 2025-2026.

1.3 Goose (Block)

Field	Value
Repo	github.com/block/goose
Stars	33,554
License	Apache-2.0
Language	Rust
Created by	Block (Square, Cash App)
Latest commit	Mar 25, 2026

What it does: An extensible AI agent from Block that goes beyond code suggestions – it can install dependencies, execute code, edit files, run tests, and orchestrate workflows. Built on the Model Context Protocol (MCP) for deep tool integration.

Architecture:



Key architectural innovations: - **MCP-native:** Built from the ground up on the Model Context Protocol (co-developed with Anthropic). Every external integration is an MCP server, making Goose infinitely extensible. - **Rust core:** Fast startup, low memory footprint, and reliable cross-platform binaries. - **Enterprise-grade:** Backed by Block (a \$40B+ company), designed for enterprise workflows from day one.

Models supported: Any LLM – supports multi-model configuration to optimize performance and cost. OpenAI, Anthropic, Google, local models, and any OpenAI-compatible endpoint.

Key features: - MCP-first architecture with plugin ecosystem - Desktop app and CLI interfaces - Multi-model configuration for cost optimization - File editing, terminal execution, test running - Workflow orchestration beyond just coding - Can interact with external APIs autonomously

Strengths: - Rust-based (fast, efficient, reliable) - MCP-native gives it the best extensibility story - Enterprise backing from Block - Free and fully open source - Active development (commits daily)

Weaknesses: - Narrower community compared to OpenCode/Aider - Still evolving rapidly (API surface changing) - Less coding-specific benchmarking data than Aider - Documentation could be more comprehensive

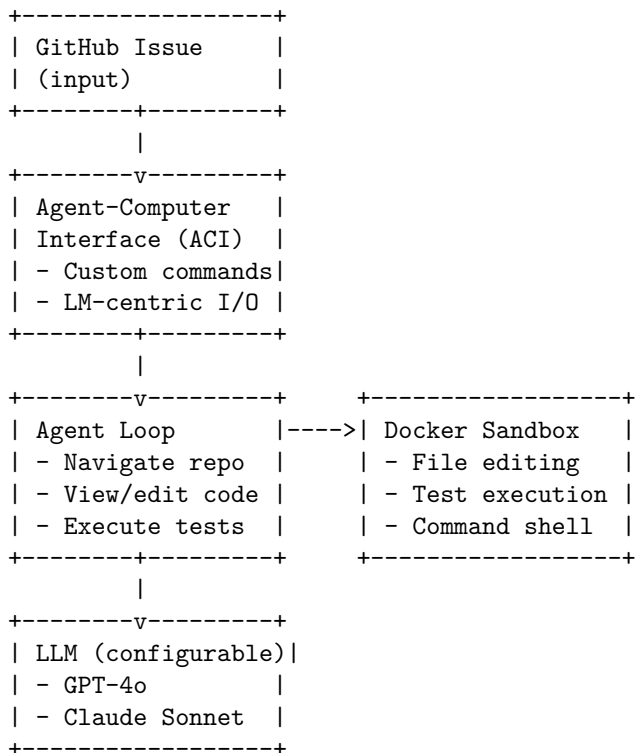
Community: 33.5K stars, 3,124 forks, 368+ contributors. Backed by Block with dedicated engineering team.

1.4 SWE-agent (Princeton)

Field	Value
Repo	github.com/princeton-nlp/SWE-agent
Stars	18,844
License	MIT
Language	Python
Created by	Princeton & Stanford researchers
Latest commit	Mar 24, 2026

What it does: An autonomous agent that takes a GitHub issue and attempts to fix it automatically. Designed primarily as a research tool for the SWE-bench benchmark, it also supports offensive cybersecurity testing and competitive coding challenges. Published at NeurIPS 2024.

Architecture:



Key architectural innovations: - **Agent-Computer Interface (ACI):** Custom-designed commands and feedback formats that make it easier for LLMs to navigate repositories, view/edit code, and execute tests. The ACI is the key insight – the interface design matters as much as the model. - **YAML-governed:** The entire agent behavior is controlled by a single YAML configuration file. - **mini-SWE-agent:** A simplified 100-line version that scores >74% on SWE-bench verified, proving the core ideas are sound.

Models supported: GPT-4o, Claude Sonnet 4, and any LLM that supports function calling. Configured per-run via YAML.

Key features: - Autonomous GitHub issue resolution - Docker-sandboxed execution - SWE-bench benchmarking built-in - Cybersecurity vulnerability finding - Competitive coding challenge solving - Configurable via single YAML file - Minimal, research-oriented design

Strengths: - Academic rigor (Princeton/Stanford, NeurIPS 2024) - State-of-the-art SWE-bench performance - Clean, well-documented architecture - The ACI concept is influential across the ecosystem - mini-SWE-agent proves simplicity works

Weaknesses: - Research-first, not production-ready for daily development - No interactive mode (batch processing only) - No MCP or plugin system - Requires Docker for sandboxing - Not designed for general-purpose coding assistance

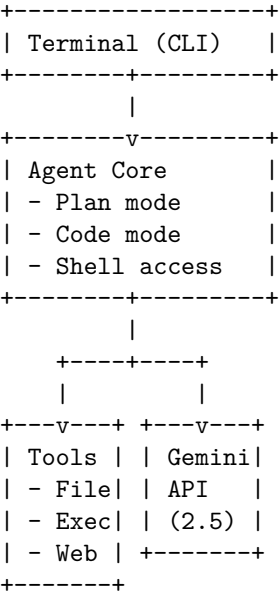
Community: 18.8K stars, 2,034 forks. Academic project with active research development.

1.5 Gemini CLI (Google)

Field	Value
Repo	github.com/google-gemini/gemini-cli
Stars	99,036
License	Apache-2.0
Language	TypeScript
Created by	Google
Latest commit	Mar 25, 2026

What it does: Google’s open-source AI agent that brings Gemini directly into the terminal. A direct competitor to Claude Code, offering agentic coding with file editing, terminal execution, and multimodal understanding powered by Gemini models with massive context windows (1M+ tokens).

Architecture:



Key features: - Free tier with generous token limits via Google AI Studio - Plan mode (shipped March 2026) for multi-step reasoning - 1M+ token context window (Gemini 2.5 Pro) - Multimodal (images, code, text) - MCP support for extensibility - AGENTS.md project configuration - Shell integration and file editing

Models supported: Gemini 2.5 Pro, Gemini 2.5 Flash, and other Gemini family models. Primarily Gemini-only (not model-agnostic).

Strengths: - Massive context window (1M+ tokens) - Free tier makes it accessible - Backed by Google with active development - Second-most-starred coding CLI (99K) - Multimodal capabilities

Weaknesses: - Gemini-only (no model flexibility) - Newer than Aider/Claude Code (less battle-tested) - Google API dependency - Community perception of Google project longevity

Community: 99K stars, 12,607 forks. Google-backed with rapid community adoption.

Note: Gemini CLI is covered in more detail in [13-gemini-cli-architecture.md](#).

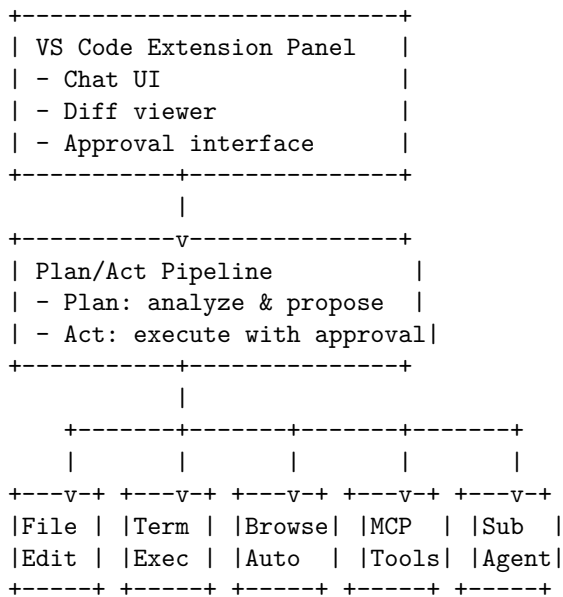
2. IDE Extensions

2.1 Cline

Field	Value
Repo	github.com/cline/cline
Stars	59,338
License	Apache-2.0
Language	TypeScript
Originally	Claude Dev (by Saoud Rizwan)
Latest commit	Mar 25, 2026

What it does: The most popular open-source autonomous coding agent for VS Code. Cline can create/edit files, execute terminal commands, use a browser, and integrate with MCP tools – all with human approval at every step.

Architecture:



Key architectural innovations: - **Plan/Act pipeline:** Every task is decomposed into a planning phase and an execution phase. Changes are sorted into snapshots that can be approved or rolled back. - **Human-in-the-loop:** Every file edit, terminal command, and browser action requires explicit user approval, making it safe for production codebases. - **Browser automation:** Built-in Puppeteer-based browser control for frontend testing – can click elements, type into fields, scroll, capture screenshots and console logs. - **MCP Marketplace:** 100+ pre-built MCP servers for database access, API integrations, browser control, and custom tools. - **Native subagents (v3.58):** Parallel execution of sub-tasks for complex workflows.

Models supported: More providers than any other tool – OpenRouter, Anthropic, OpenAI, Google Gemini, AWS Bedrock, Azure, GCP Vertex, Cerebras, Groq, and local models via LM Studio or Ollama. Any OpenAI-compatible API.

Key features: - Autonomous file creation/editing with diff view - Terminal command execution - Browser automation (click, type, scroll, screenshot) - MCP tool integration with marketplace (100+ servers) - Native subagents for parallel execution (v3.58) - CLI 2.0 with headless CI/CD mode (Feb 2026) - Context injection via @url, @problems, @file, @folder - Per-task token and cost tracking - .clinerules files for project-specific configuration - 5M+ installs across VS Code, Cursor, JetBrains, Zed, Neovim

Strengths: - Most-installed open-source coding extension (5M+) - Best browser automation of any coding tool - Richest MCP ecosystem - Model-agnostic with the widest provider support - Human-in-the-loop safety by default - Rapidly shipping features (subagents, CLI, etc.)

Weaknesses: - Can be expensive with token usage (long system prompts) - Approval fatigue on large tasks (many confirmation dialogs) - VS Code-dependent for the extension (CLI is newer) - Context window management with local models requires tuning

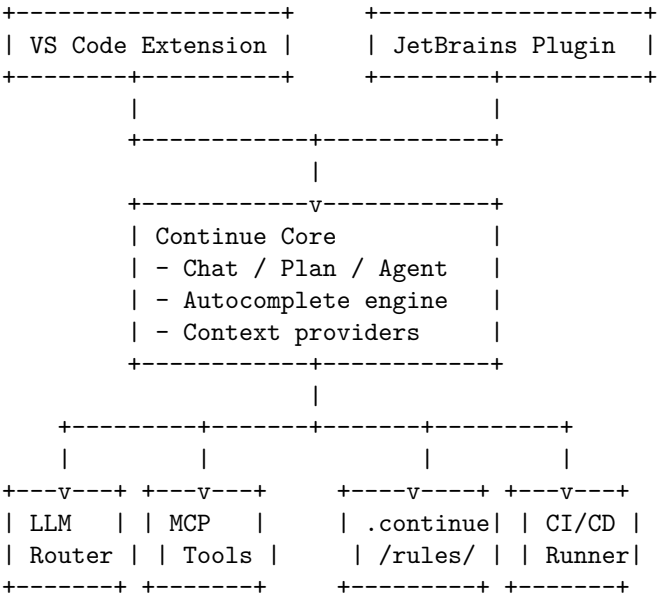
Community: 59.3K stars, 6,027 forks, 698 open issues. One of the fastest-growing VS Code extensions ever.

2.2 Continue.dev

Field	Value
Repo	github.com/continuedev/continue
Stars	32,047
License	Apache-2.0
Language	TypeScript
Created by	Continue (YC-backed)
Latest commit	Mar 25, 2026

What it does: An open-source AI code assistant for VS Code and JetBrains that emphasizes model flexibility, team configuration, and CI/CD integration. Positions itself as “quality control for your software factory” with source-controlled AI checks enforceable in CI.

Architecture:



Key architectural innovations: - **Source-controlled AI rules:** `.continue/rules/` directory stores team standards, coding patterns, and AI behaviors. Shared via git, ensuring every team member gets the same AI assistant configuration. - **CI/CD integration:** Continue CLI can run AI checks in GitHub Actions, Jenkins, GitLab CI, and cron jobs – making AI-powered code review enforceable. - **Three interaction modes:** Chat (conversational), Plan (multi-step), Agent (autonomous execution).

Models supported: Fully model-agnostic – any cloud provider (OpenAI, Anthropic, Google, etc.) or local model (Ollama, LM Studio, CodeLlama, Mistral, Llama). BYOK (Bring Your Own Key) model.

Key features: - Chat, Plan, and Agent modes - Tab autocomplete with any model - MCP tool support (GitHub, Sentry, Snyk, Linear) - `.continue/rules/` for team-shared AI configuration - CI/CD pipeline integration via Continue CLI - Context providers for codebase awareness - JetBrains support (not just VS Code) - Codebase indexing for retrieval

Strengths: - Best IDE breadth (VS Code + full JetBrains family) - CI/CD integration is unique in the space - Team-oriented configuration management - Fully model-agnostic with BYOK - Apache-2.0 license with no vendor lock-in - YC-backed with sustainable business model

Weaknesses: - Less autonomous than Cline (more assistant than agent) - No browser automation - Autocomplete quality depends heavily on model choice - Configuration can be complex for beginners - Extension UX not as polished as Cline/Roo Code

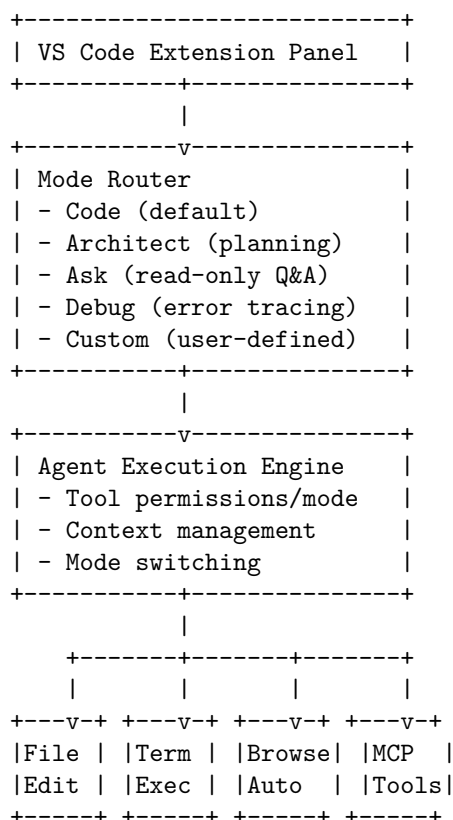
Community: 32K stars, 4,296 forks, 528 open issues. Strong enterprise adoption.

2.3 Roo Code

Field	Value
Repo	github.com/RooVetGit/Roo-Code
Stars	22,830
License	Apache-2.0
Language	TypeScript
Originally	Roo-Cline (fork of Cline)

What it does: A VS Code extension that provides a team of AI agents with specialized modes for different development tasks – planning, coding, debugging, architecture, and custom workflows. Think of it as Cline with a mode system.

Architecture:



Key architectural innovations:

- **Mode system:** Five built-in modes (Code, Architect, Ask, Debug, Custom) each with scoped tool permissions. Modes keep models focused and limit context window pollution.
- **Mode Gallery:** Community-published mode configurations for common workflows (backend scaffolding, CI/CD editing, test generation).
- **Smart mode switching:** Modes can detect when a task falls outside their scope and suggest switching to the appropriate mode.

Models supported: Model-agnostic – OpenAI, Anthropic, Google, local models via Ollama, and any other provider. BYOK.

Key features:

- 5 built-in modes + custom mode templates
- Mode Gallery for community-shared configurations
- Multi-file editing, terminal execution, browser automation
- MCP tool integration
- Intelligent task coordination across modes
- Model-agnostic architecture
- **.roorules** for project configuration
- v3.50.4 (Feb 2026)

Strengths:

- Mode system is genuinely innovative for task separation
- Forked from Cline, so inherits all its capabilities
- Active community with mode marketplace
- Good for teams wanting role-specific AI agents
- Rapid release cadence

Weaknesses:

- Smaller community than Cline (the project it forked from)
- Mode switching can add cognitive overhead
- Some features lag behind Cline's latest releases
- Documentation still maturing
- Forked identity can cause confusion with Cline

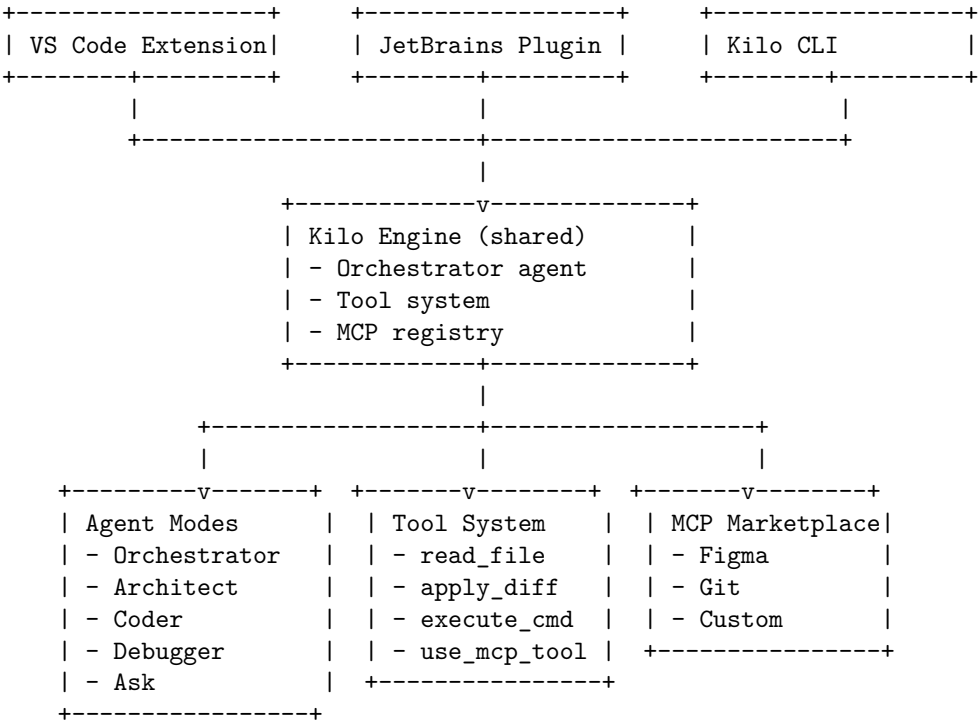
Community: 22.8K stars, 2,934 forks, 743 open issues. Fastest-evolving AI coding extension.

2.4 Kilo Code

Field	Value
Repo	github.com/Kilo-Org/kilocode
Stars	17,177
License	MIT
Language	TypeScript
Created by	Kilo AI
Latest commit	Mar 25, 2026

What it does: An all-in-one agentic engineering platform for VS Code, JetBrains, and CLI. Claims to be the #1 coding agent on OpenRouter with 1.5M+ users and 25T+ tokens processed.

Architecture:



Key architectural innovations: - **Shared engine:** One engine powers VS Code extension, JetBrains plugin, and CLI. Same capabilities everywhere. - **Orchestrator mode:** A meta-agent that plans and delegates to Architect, Coder, and Debugger agents. - **Parallel agent execution:** Can run multiple agents simultaneously for multi-step tasks. - **MCP Marketplace:** Curated collection of Skills, MCP Servers, and Modes for the Kilo ecosystem.

Models supported: All major providers + local models. Notable partnership with xAI for “Grok Code Fast” with unlimited free access and 262K token context.

Key features: - 5 agent modes (Orchestrator, Architect, Coder, Debugger, Ask) - Parallel agent execution - MCP Marketplace with Skills and Modes - Inline autocomplete - Browser automation - Streamable HTTP MCP support (April 2025) - VS Code + JetBrains + CLI - 1.5M+ users, 25T+ tokens processed

Strengths: - Multi-platform (VS Code, JetBrains, CLI) - Orchestrator mode for complex multi-step tasks - Large user base (1.5M+) - Active MCP marketplace - MIT license (most permissive)

Weaknesses: - Newer project, less battle-tested than Cline - Heavy marketing claims (#1 on OpenRouter) may not reflect all use cases - Community size smaller than Cline/Roo Code by stars - Documentation could be more technical

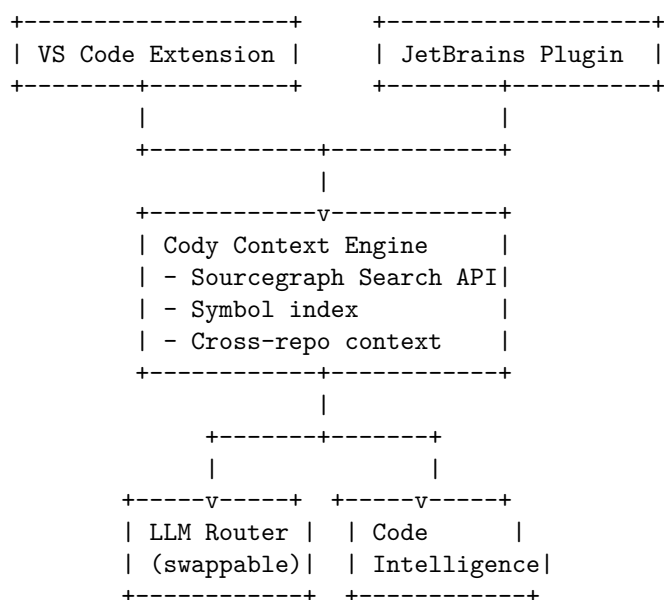
Community: 17.2K stars, 2,223 forks. Growing rapidly with dedicated marketplace.

2.5 Cody (Sourcegraph)

Field	Value
Repo	github.com/sourcegraph/cody-public-snapshot
Stars	3,794 (public snapshot)
License	Apache-2.0
Language	TypeScript
Created by	Sourcegraph
Status	Pivoted to “Amp” (enterprise)

What it does: An AI coding assistant that leverages Sourcegraph’s deep code search and intelligence to provide context-aware code assistance. Uniquely strong at understanding large, enterprise-scale codebases due to Sourcegraph’s decade of code indexing technology.

Architecture:



Key features: - Deep codebase context via Sourcegraph’s code intelligence - Autocomplete (single and multi-line) - Chat with repository-wide context - Inline edit and refactor - Swappable LLMs (Claude Sonnet, GPT-4o, Gemini) - Cross-repository context awareness

Current status: Public free/pro plans discontinued in 2025. Cody has been rebranded as **Amp** for enterprise use. The open-source public snapshot is a frozen copy. This is now primarily an enterprise product, not a community open-source tool.

Community: 3.8K stars on public snapshot. Enterprise-focused going forward.

3. Full IDEs / Self-Hosted Platforms

3.1 Void

Field	Value
Repo	github.com/voideditor/void
Stars	28,469
License	Apache-2.0
Language	TypeScript
Created by	Void (YC-backed)

Field	Value
Latest commit	Jan 12, 2026
Status	PAUSED

What it does: An open-source, privacy-focused AI code editor built as a fork of VS Code. Positioned as a free alternative to Cursor with full model flexibility and local execution.

Architecture:

```

+-----+
| VS Code Fork (Electron) |
| +-----+ |
| | Agent Mode | Gather Mode | |
| | Chat Panel | Inline Editor | |
| +-----+ |
| | | | |
| +-----+ |
| | LLM Streaming Layer | |
| | - Real-time UI updates | |
| | - Multiple provider support | |
| +-----+ |
| | File Indexing Engine | |
| +-----+ |
+-----+

```

Key features: - Agent Mode, Gather Mode, and Chat - AI-powered code completion and inline editing - Checkpoint and visualize changes - File indexing for efficient navigation - Direct LLM connections (no data retention) - Support for Claude, GPT, Gemini, Ollama

Current status: Development paused. The team announced they are “exploring novel coding ideas” and while Void will continue running, existing features may stop working without maintenance. Despite 28K+ stars, this project is not actively maintained as of early 2026.

Community: 28.5K stars, 2,375 forks. High interest but paused development is a significant risk.

3.2 Pear AI

Field	Value
Repo	github.com/trypear/pearai-app
Stars	670
License	MIT
Language	TypeScript
Created by	Pear (YC-backed)
Latest commit	May 2025
Status	Low activity

What it does: An open-source AI code editor (VS Code fork) that bundles multiple AI tools – Aider for code generation, Supermaven for autocomplete, Continue for assistant features, Mem0 for memory, and Perplexity for search – into a single editor experience.

Architecture:

```

+-----+
| VS Code Fork (Electron) |
| +-----+ |
| | PearAI Router (model select) | |

```

```

| +-----+
| | Integrated Tools: |
| | - Aider (code gen) |
| | - Supermaven (autocomplete) |
| | - Continue (assistant) |
| | - Roo Code/Cline (agent) |
| | - Mem0 (memory) |
| | - Perplexity (search) |
| +-----+
+-----+

```

Key features: - Bundles best-of-breed OSS tools into one editor - PearAI Router for intelligent model selection - AI Chat, inline prompts, agent mode - Context-aware with local codebase indexing - Supported models: GPT-4, Claude, Llama

Current status: Low activity. Last commit May 2025. With only 670 stars, this project has not gained significant traction. The “bundle existing tools” approach means it depends on upstream projects that are evolving fast.

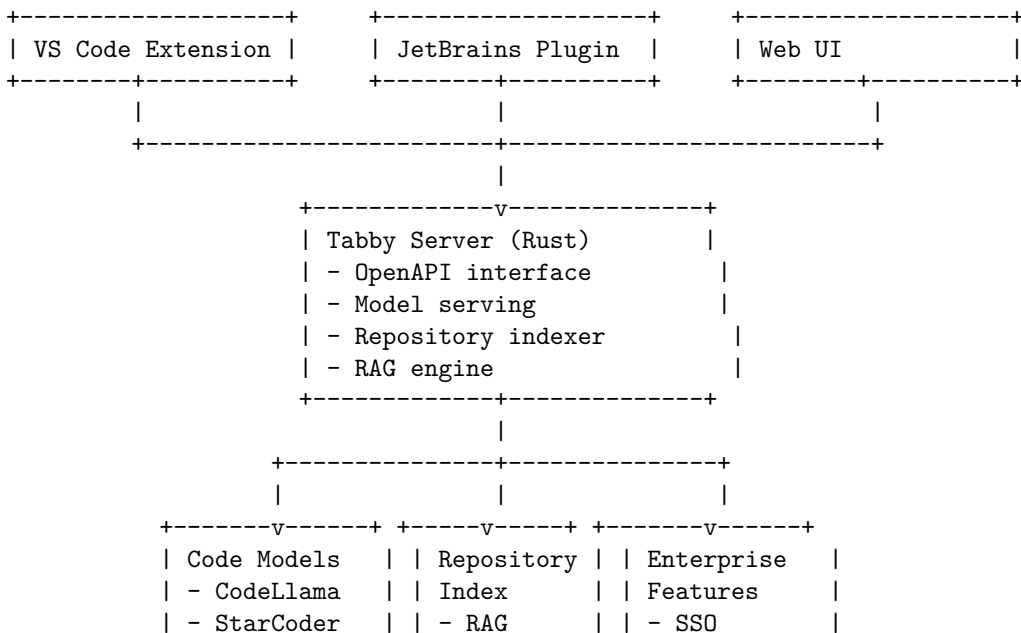
Community: 670 stars, 190 forks. YC-backed but struggling for adoption.

3.3 Tabby

Field	Value
Repo	github.com/TabbyML/tabby
Stars	33,050
License	Apache-2.0
Language	Rust
Created by	TabbyML
Latest commit	Mar 2, 2026

What it does: A self-hosted AI coding assistant, offering an on-premises alternative to GitHub Copilot. Runs on consumer-grade GPUs with no cloud dependency, providing real-time code completion, chat, and repository-level understanding.

Architecture:



- CodeGen	- Context	- Analytics	
- Custom	+-----+	- Teams	
+-----+		+-----+	

Key architectural innovations: - **Self-contained server:** No external DBMS or cloud service needed. Everything runs locally with an OpenAPI interface for easy integration. - **Rust-based server:** Fast, low-memory, reliable. Runs on consumer GPUs (NVIDIA, AMD). - **Repository-level RAG:** Indexes your entire repository for context-aware completions and chat. Can index GitLab merge requests for additional context. - **Enterprise-ready:** Team management, usage analytics, SSO, fine-tuning on private repos.

Models supported: Self-hosted coding models – CodeLlama, StarCoder, CodeGen, DeepSeek Coder, and custom fine-tuned models. Not designed for cloud API models.

Key features: - Real-time code completion (40+ languages) - Chat interface with repository context - Repository indexing with RAG - Fine-tuning on private repositories - Consumer-grade GPU support - OpenAPI interface for integration - Enterprise: SSO, team management, analytics - VS Code and JetBrains extensions - Pages feature for persistent, shareable answers

Strengths: - Best self-hosted option (no cloud dependency) - Rust-based (fast, efficient) - Enterprise features (SSO, analytics, teams) - Privacy-first (all data stays on-prem) - Repository-level context awareness - Active development with regular releases

Weaknesses: - Requires GPU hardware for good performance - Smaller model ecosystem than cloud-based tools - No agent/autonomous capabilities (completion + chat only) - No MCP or plugin system - Self-hosting complexity (GPU drivers, model management)

Community: 33K stars, 1,692 forks. Strong enterprise and privacy-focused community.

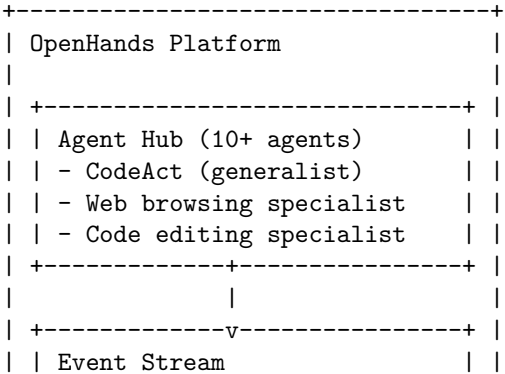
4. Agent Platforms

4.1 OpenHands

Field	Value
Repo	github.com/All-Hands-AI/OpenHands
Stars	69,738
License	MIT
Language	Python
Originally	OpenDevin
Latest commit	Mar 25, 2026

What it does: An open platform for AI software developers as generalist agents. OpenHands agents can write code, use the command line, browse the web, and interact with the world in ways similar to a human developer. Published at ICLR 2025.

Architecture:



		- Chronological actions		
		- Observations		
		- User interactions		
	+	-----+	+	
	+	-----v-----+	+	
		Runtime (Docker Sandbox)		
		- Bash shell		
		- Web browser		
		- IPython server		
		- File system		
	+	-----+	+	
	+	-----+	+	
		SDK (Python library)		
		- Define agents in code		
		- Run locally or cloud		
		- Scale to 1000s of agents		
	+	-----+	+	
+	+	-----+	+	

Key architectural innovations: - **3-layer architecture:** Agent abstraction (pluggable agents), Event stream (action/observation history), Runtime (Docker-sandboxed execution). - **CodeAct agent:** A strong generalist agent based on the CodeAct architecture that uses code execution as its primary action mechanism, with additions for web browsing and code editing. - **Composable SDK:** Python library that lets you define agents in code, run locally, or scale to thousands of agents in the cloud. - **Event stream:** Chronological tracking of all actions and observations, enabling replay, debugging, and audit trails.

Models supported: Model-agnostic – works with any LLM including OpenAI, Anthropic, Google, local models. Designed to be provider-independent.

Key features: - 10+ implemented agents in the agent hub - Docker-sandboxed runtime with bash, browser, IPython - Event stream for full audit trail - Python SDK for custom agent development - Web UI for interactive use - Cloud scaling (1000s of agents) - SWE-bench evaluation built-in - AMD Lemonade Server integration for local AI

Strengths: - Largest AI agent platform by stars (70K) - Academic rigor (ICLR 2025, arXiv papers) - Most comprehensive agent framework - Docker sandboxing for safety - SDK enables building custom agents - Active community (188+ contributors, 2.1K+ contributions)

Weaknesses: - Heavy infrastructure requirements (Docker, Python) - Steep learning curve for custom agent development - Not a simple “coding assistant” – it’s a platform - Overkill for simple code editing tasks - Resource-intensive (Docker containers per session)

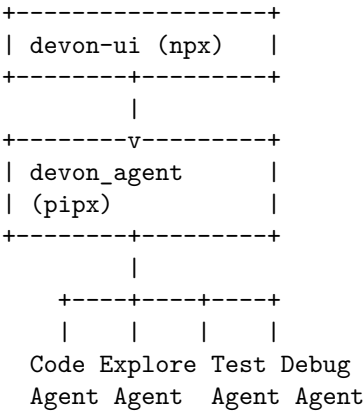
Community: 69.7K stars, 8,741 forks. Largest open-source AI agent platform. Published research papers. Very active development.

4.2 Devon

Field	Value
Repo	github.com/entropy-research/Devon
Stars	3,454
License	AGPL-3.0
Language	Python
Created by	Entropy Research
Latest commit	Jul 2024
Status	STALE

What it does: An open-source pair programmer with multi-agent architecture. Devon deploys specialized agents for code generation, exploration, configuration writing, testing, and bug fixing. Featured on Hacker News in May 2024.

Architecture:



Key features: - Multi-file editing - Codebase exploration - Test writing and bug fixing - Configuration writing - Multi-agent architecture

Models supported: Anthropic, OpenAI, Groq APIs.

Current status: Stale. Last commit July 2024. No activity for 20+ months. The project showed early promise but appears abandoned. The AGPL-3.0 license is also more restrictive than alternatives.

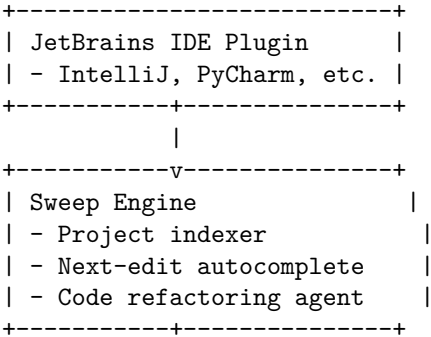
Community: 3.5K stars, 279 forks. Initial HN buzz but no sustained development.

4.3 Sweep AI

Field	Value
Repo	github.com/sweepai/sweep
Stars	7,655
License	Proprietary (OSS core)
Language	Python/Jupyter
Created by	Sweep AI
Latest commit	Sep 2025
Status	Pivoted to JetBrains

What it does: Originally an AI junior developer that automatically resolves GitHub issues by reading your codebase and creating pull requests. Has since pivoted to become a JetBrains IDE plugin focused on autocomplete, inline editing, and agent capabilities.

Architecture (current JetBrains focus):



```

      |
+-----v-----+
| Sweep's Own LLMs |
| (no third-party retention) |
+-----+

```

Key features: - Next-edit autocomplete (unique to Sweep) - Context-aware inline suggestions - Automated refactoring (function extraction, dead code removal) - Test generation - Static analysis feedback - Uses own LLMs (no code retained by third parties) - Supports Python, JS/TS, Java, Go, C#, C++, Rust

Current status: Active as a JetBrains plugin but the open-source GitHub issue resolver (the original product) has been deprioritized. The GitHub repo is primarily for the legacy product.

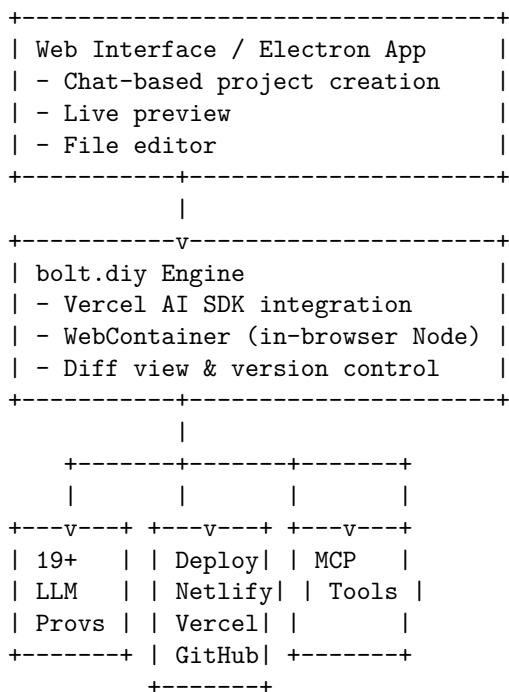
Community: 7.7K stars, 454 forks. 40K+ JetBrains installs.

4.4 bolt.diy

Field	Value
Repo	github.com/stackblitz-labs/bolt.diy
Stars	19,204
License	MIT
Language	TypeScript
Created by	Community (forked from bolt.new)
Latest commit	Feb 5, 2026

What it does: An open-source platform for prompting, running, editing, and deploying full-stack web applications using any LLM. Community fork of StackBlitz's bolt.new that adds multi-provider LLM support.

Architecture:



Key architectural innovations: - **WebContainer:** Runs a full Node.js environment in the browser via StackBlitz's WebContainer technology. No server needed for previews. - **Multi-provider LLM support:** The key differentiator from bolt.new – supports 19+ providers including OpenAI, Anthropic, Ollama, OpenRouter, Gemini, Mistral, DeepSeek, Groq, and more.

Models supported: 19+ providers – OpenAI, Anthropic, Ollama, OpenRouter, Google Gemini, LM Studio, Mistral, xAI, HuggingFace, DeepSeek, Groq, Cohere, Together AI, Perplexity, Amazon Bedrock, and more. Extensible via Vercel AI SDK.

Key features: - Full-stack web app generation from prompts - In-browser live preview (WebContainer) - 19+ LLM providers - Deploy to Netlify, Vercel, or GitHub Pages - Image attachment for context - Version history and revert - Diff view for AI changes - MCP support - Docker deployment option - Electron desktop app - Download as ZIP

Strengths: - Best tool for full-stack web app generation - In-browser execution (no Docker/server needed) - Widest LLM provider support for web app builders - One-click deployment to major platforms - Active community fork with rapid features

Weaknesses: - Web-app focused (not general-purpose coding) - Generated code quality varies by model - No terminal/CLI access (browser-only) - WebContainer limitations for non-web projects - Community-maintained (no corporate backing)

Community: 19.2K stars, 10,383 forks. Very active community fork. bolt.new (proprietary) has 16.3K stars separately.

5. Archived / Inactive

5.1 Mentat

Field	Value
Repo	github.com/AbanteAI/archive-old-cli-mentat
Stars	2,559
License	Apache-2.0
Language	Python
Status	ARCHIVED

What it was: A CLI-based AI coding assistant that coordinated edits across multiple files and locations. Distinguished itself from Copilot by handling multi-file edits and from ChatGPT by automatically having project context.

Current status: The original CLI has been archived. AbanteAI pivoted Mentat into an AI-powered GitHub bot (mentat.ai) that writes and reviews code as a SaaS product.

5.2 CodeGeeX

Field	Value
Repo	github.com/THUDM/CodeGeeX4
Stars	2,455 (CodeGeeX4) / 7,593 (CodeGeeX2)
License	Apache-2.0
Language	Python
Created by	Tsinghua University
Status	Low activity (model weights, not tool)

What it is: A multilingual code generation model (13B parameters) developed by Tsinghua University. Provides code completion, translation, and generation across 20+ programming languages. Available via IDE plugins and self-hosted deployment.

Key specs: 128K token context window, 82.3% on HumanEval (CodeGeeX4-ALL-9B). This is more of a model release than a tool – the IDE plugins wrap the model.

Ecosystem Analysis

Growth Trajectory (2024 -> 2026)

The open-source AI coding tool ecosystem has exploded:

Metric	2024	2026
Total stars (top 10 tools)	~50K	~550K+
Tools with MCP support	0	8+
Tools with browser automation	1	5+
CLI tools viable for daily use	2	6+
Tools with subagent/parallel exec	0	4+

Key Trends

- MCP is becoming table stakes.** Tools without MCP support (Aider, SWE-agent) are feeling the pressure. MCP enables plugin ecosystems that dramatically extend capabilities.
- CLI resurgence.** After years of IDE-heavy development, the terminal is back as the center of gravity for AI-assisted coding. OpenCode (130K stars), Gemini CLI (99K), and Aider (42K) prove this.
- Mode/Agent systems.** The shift from “one AI assistant” to “a team of specialized agents” is happening across Roo Code, Kilo Code, and OpenHands. Expect every tool to ship modes by end of 2026.
- Persistent sessions.** OpenCode’s persistent background server is a game-changer. Sessions that survive disconnects are essential for long-running tasks.
- CI/CD integration.** Continue.dev’s CI/CD runner points to the future – AI checks enforced in pipelines alongside linting and tests.
- Self-hosting demand.** Tabby’s 33K stars show strong demand for on-premises solutions, especially in enterprise and regulated industries.
- Convergence of interfaces.** Tools are no longer just CLI or just IDE extension. OpenCode, Kilo Code, and Goose all ship CLI + IDE + desktop apps from a single engine.

The Fork Tree

Understanding the lineage helps explain feature overlap:

```
VS Code (Microsoft)
|
+-- Cursor (proprietary)
+-- Void (fork, paused)
+-- Pear AI (fork, low activity)

Cline (Claude Dev)
|
+-- Roo Code (fork, active)
+-- Kilo Code (fork lineage, now independent)

bolt.new (StackBlitz)
|
+-- bolt.diy (community fork)

OpenDevin
|
+-- OpenHands (renamed, thriving)
```

Recommendations by Use Case

“I want a CLI tool for daily coding”

1. **Aider** – Most mature, best git integration, widest model support
2. **OpenCode** – Most features, persistent sessions, largest community
3. **Goose** – Best MCP integration, Rust-fast, enterprise-backed

“I want an AI agent in my VS Code”

1. **Cline** – Most installs, best browser automation, richest MCP ecosystem
2. **Roo Code** – Best mode system for task separation
3. **Continue.dev** – Best for teams (shared rules, CI/CD integration)
4. **Kilo Code** – Best multi-platform (VS Code + JetBrains + CLI)

“I want to self-host everything on-prem”

1. **Tabby** – Purpose-built for self-hosting, runs on consumer GPUs
2. **OpenCode + Ollama** – Full agent with local models
3. **Continue.dev + Ollama** – IDE assistant with local models

“I want autonomous agents that solve GitHub issues”

1. **OpenHands** – Most comprehensive agent platform
2. **SWE-agent** – Best benchmarked, academic rigor
3. **Sweep AI** – Original issue-to-PR workflow (now JetBrains-focused)

“I want to generate full web apps from prompts”

1. **bolt.diy** – Best multi-provider support, in-browser execution

“I want the biggest community and ecosystem”

1. **OpenCode** (130K stars)
2. **Gemini CLI** (99K stars)
3. **OpenHands** (70K stars)
4. **Cline** (59K stars)

This document covers the open-source AI coding tool landscape as of March 25, 2026. The ecosystem is evolving rapidly – star counts and feature sets may change significantly within weeks. All star counts and commit dates were verified via the GitHub API at the time of writing.